FLIPKART SALES DASHBOARD REPORT

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1. ABSTRACT

This project focuses on analyzing Flipkart sales data using Python and Power BI to generate meaningful business insights. The dataset includes information about products, categories, prices, discounts, ratings, and customer orders.

The main objective of this project is to understand sales performance, identify top-selling products, and analyze customer preferences. The dashboard highlights key metrics such as total revenue, total orders, average order value, and total quantity sold.

This project demonstrates how data visualization helps convert raw e-commerce data into actionable insights for better decision-making.

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2. INTRODUCTION

E-commerce platforms like Flipkart generate large volumes of data related to products, sales, and customers. Analyzing this data helps businesses improve their strategies and increase profitability.

This project aims to analyze Flipkart sales data and create an interactive Power BI dashboard. Python is used for data cleaning and preprocessing, while Power BI is used for visualization.

The dashboard provides a clear overview of sales performance and allows users to explore data using filters and slicers.

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3. OBJECTIVES

- To clean and preprocess sales data
- To analyze product and category performance
- To identify top-selling products
- To analyze customer buying patterns
- To create an interactive Power BI dashboard
- To generate business insights

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4. DATA PREPROCESSING

The dataset is processed using Python (Pandas) with the following steps:

- Removing duplicate records
- Handling missing values
- Data formatting and structuring
- Creating new column: **Total Sales = Price × Quantity**

These steps ensure accurate and consistent data for analysis.

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5. EXPLORATORY DATA ANALYSIS

5.1 Top Selling Products

Identifies products with highest sales quantity and revenue

5.2 Category-wise Sales

Analyzes which categories generate the most revenue

5.3 Discount Analysis

Studies impact of discounts on sales

5.4 Customer Ratings

Analyzes product ratings and customer satisfaction

5.5 Order Distribution

Examines how orders vary across categories and regions

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6. POWER BI DASHBOARD

6.1 Key Performance Indicators (KPIs)

- Total Revenue
- Total Orders
- Average Order Value
- Total Quantity Sold

6.2 Visualizations Used

- **Bar Chart**: Top selling products
- **Pie/Donut Chart**: Category-wise sales distribution

- **Line Chart**: Sales trends over time
- **Column Chart**: Discount vs sales analysis
- **Bar Chart**: Ratings distribution
- **Table**: Product details and sales
- **Slicers**: Category, product, and rating filters

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7. KEY INSIGHTS

- Few products contribute major portion of sales
- Electronics and fashion categories dominate revenue
- Discounts increase sales volume
- Higher-rated products attract more customers
- Customer preferences vary across categories

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8. ADVANTAGES

- Easy to understand visualizations
- Interactive dashboard with filters
- Helps identify top-performing products
- Supports business decision-making
- Scalable for large datasets

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9. LIMITATIONS

- Limited real-time data
- No predictive analytics
- Data accuracy depends on source
- Limited customer-level insights

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10. FUTURE ENHANCEMENTS

- Add real-time data integration

- Implement sales prediction using ML
- Add customer segmentation
- Improve dashboard interactivity
- Deploy on cloud platform

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11. CONCLUSION

The Flipkart Sales Dashboard successfully demonstrates how Python and Power BI can be used to analyze e-commerce data effectively.

The dashboard provides valuable insights into product performance, customer behavior, and sales trends. With further enhancements, it can become a powerful business intelligence tool for e-commerce platforms.